Capability Maturity Model (CMM)

Introduction

Capability Maturity Model (CMM) is a process improvement framework developed by the Software Engineering Institute (SEI) at Carnegie Mellon University. It helps organizations assess the maturity of their software development processes and provides a structured roadmap for continuous improvement, higher product quality, and predictable project outcomes.

Objectives

The primary objectives are to improve software quality, standardize development processes, reduce project risks, increase productivity, improve customer satisfaction, enhance project planning, and establish a culture of continuous process improvement.

Types of CMM

SW-CMM: Software development.
P-CMM: Workforce capability.
SA-CMM: Software acquisition.
SE-CMM: Systems engineering.

Five Maturity Levels

Level 1 - Initial
Processes are informal, unpredictable, and depend on individual efforts. Success is inconsistent.

Level 2 - Managed
Projects follow basic planning, tracking, requirement management, and quality assurance practices.

Level 3 - Defined
Organization-wide standard processes are documented, standardized, and consistently followed.

Level 4 - Quantitatively Managed
Performance is measured using metrics. Decisions are based on statistical analysis and process data.

Level 5 - Optimizing
Focuses on innovation, defect prevention, continuous improvement, and adopting new technologies.

Advantages

Improves software quality, increases productivity, reduces defects, enhances customer confidence, supports better estimation, lowers development cost, and creates repeatable and reliable development practices.

Limitations

Implementation requires time, cost, documentation, training, management commitment, and may be difficult for small organizations.

Applications

Widely used in software development companies, banking, healthcare, government projects, telecom, aerospace, manufacturing, and IT service organizations.

Difference Between CMM and CMMI

Feature	CMM	CMMI
Purpose	Software process improvement	Enterprise-wide process improvement
Scope	Single model	Integrated model
Focus	Software development	Software, services & systems
Approach	Fixed maturity model	Staged & Continuous
Current Status	Older framework	Modern industry standard

Conclusion

CMM provides organizations with a systematic approach to process maturity. By progressing through the five maturity levels, organizations achieve better quality, predictable schedules, reduced risks, and continuous improvement. Although CMM has largely been replaced by CMMI, its principles remain the foundation of modern process improvement models.